METRIC
A-A-54813
12 February 1993

COMMERCIAL ITEM DESCRIPTION

RESIN, ACRYLIC, DENTAL, FIBERED, Denture Base Material

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for Military Specification MIL-R-37770A which is cancelled.

This Commercial Item Description covers a fibered acrylic base resin (polymer) suitable for use in the fabrication of fibered dentures

Salient characteristics:

Except as noted below the acrylic resin shall be in accordance with the American National Standards Institute (ANSI)/American Dental Association (ADA) No. 12. In case of a conflict, this document shall govern.

Classification. The resin covered by this specification shall be Class I normal impact (power and liquid) and of the following types.

Type A - Light pink
Type B - Reddish pink

Material. The liquid shall be a mixture of methyl methacrylate, etylene glycol dimethacrylate and a storage polymerization inhibitor. The liquid shall be clear, colorless and free from deposits or sediment.

The normal impact powder shall be composed of polyacrylic acid esters and poly (substituted) acrylic acid esters or copolymers of either or a mixture of polymers and copolymers The powder shall be readily plasticized to a moldable dough when mixed with the liquid. Cellulose acetate fibers shall be distributed throughout the powder. The fibers shall be 3/8 inch ± 1/32 inch in length and shall be 7 to 9 denier.

AMSC N/A FSC 6520
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:38Z
Check the source to verify that this is the current version before use.

Type A powder fibers shall be red and violet and each pouch shall contain approximately 4000 filaments with a fiber weight of 0.025 to 0.035 grams per pouch.

Type B powder fibers shall be red and violet and each pouch shall contain approximately 10,200 filaments with a fiber weight of 0.075 to 0.085 grams per pouch.
The gingival tone powder shall be a heat-curing polymer.

General. The dental resin shall include 120 milliliters (minimum) of liquid monomer, 10 pouches of powder with each pouch containing not less than 21 grams of powder and 7 grams (minimum).

Color. The color of the processed resin shall conform to the color reflectance values specified below when tested as specified herein.

	Rd	a	b
Type A	17.8 - 19.8	17.8 - 19.8	5.5 - 7.0
Type B	17.5 - 19.5	23.8 - 25.8	6.0 - 8.0

Color stability. The color change shall be no greater than 2.0 delta E units when tested in accordance with ADA No. 12.

Certification acceptability. The resin shall be certified or acceptable by the American Dental Association. At time of award of contract, the manufacturer shall furnish, upon request, a certificate and data showing that the brand of material to be furnished is the same in quality and brand as that appearing on the list of acceptable or certified materials. Compliance with this certification requirement does not absolve the manufacturer from complete compliance with the other requirements of this specification.

Finish The processed resin shall have a smooth surface with a high gloss when polished by standard dental methods.

Workmanship. The resin shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Package (PG) One package consisting of one bottle of liquid monomer, ten pouches of polymer powder, one dispensing container of gingival tone polymer together with instructions for use as specified, constitutes one unit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:38Z
Check the source to verify that this is the current version before use.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after the delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Examination. Examination shall be conducted to determine compliance with specification requirements.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 2.5 (percent defective) and an inspection level of II. Where feasible, the same sample shall be used for the determination of two or more inspection characteristics. The unit of product for inspection purposes shall be one unit of issue.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 with an AQL of 1.0 (percent defective) at an inspection level of S-2. The unit of product for test purposes shall be one unit of issue.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:38Z
Check the source to verify that this is the current version before use.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commerical Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:38Z
Check the source to verify that this is the current version before use.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be as specified in the contract and/or order.

NOTE: The following National Stock Numbers are covered by this document:

National Stock Numbers	Item Identifications	Types
6520-00-420-5750	RESIN, ACRYLIC, DENTAL, Fibered, Denture Base Material	B
6520-00-935-4014	RESIN, ACRYLIC, DENTAL, Fibered, Denture Base Material	A

MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force-03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
USPHS
FDA-MPQAS

Project No. 6520-2415
Location: ML Disk #1/RESIN

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:38Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1 The preparing activity must complete blocks 1, 2, 3, and 8 In block 1, both the document number and revision letter should be given

2 The submitter of this form must complete blocks 4, 5, 6, and 7

3 The preparing activity must provide a reply within 30 days from receipt of the form

NOTE This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER A-A-54813	2 DOCUMENT DATE (YYMMDD) 12 February 1993

3 DOCUMENT TITLE

RESIN, ACRYLIC, DENTAL, FIBERED, Denture Base Material

4 NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed.)

5 REASON FOR RECOMMENDATION

6 SUBMITTER	
a NAME (Last, First, Middle Initial)	b ORGANIZATION
c ADDRESS (Include Zip Code)	d TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7 DATE SUBMITTED (YYMMDD)

8 PREPARING ACTIVITY	
a NAME Defense Personnel Support Center ATTN: DPSC-MSE	b TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (215)7372870 444-2870
c ADDRESS (Include Zip Code) 2800 South 20th Street Philadelphia, PA 19145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike Suite 1403 Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 *Previous editions are obsolete* 198/290

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:38Z
Check the source to verify that this is the current version before use.